FEA & Design Optimisation of a Structural Steel Bracket:

Overview:

A structural steel bracket, bolted to a fixed plate and carrying an actuator arm was required to withstand a cyclic load of ±10kN (stress ratio, R = -1) applied at a pin located 200mm away from the fixed end. This is shown in the image below, where *F* represents the cyclic load:

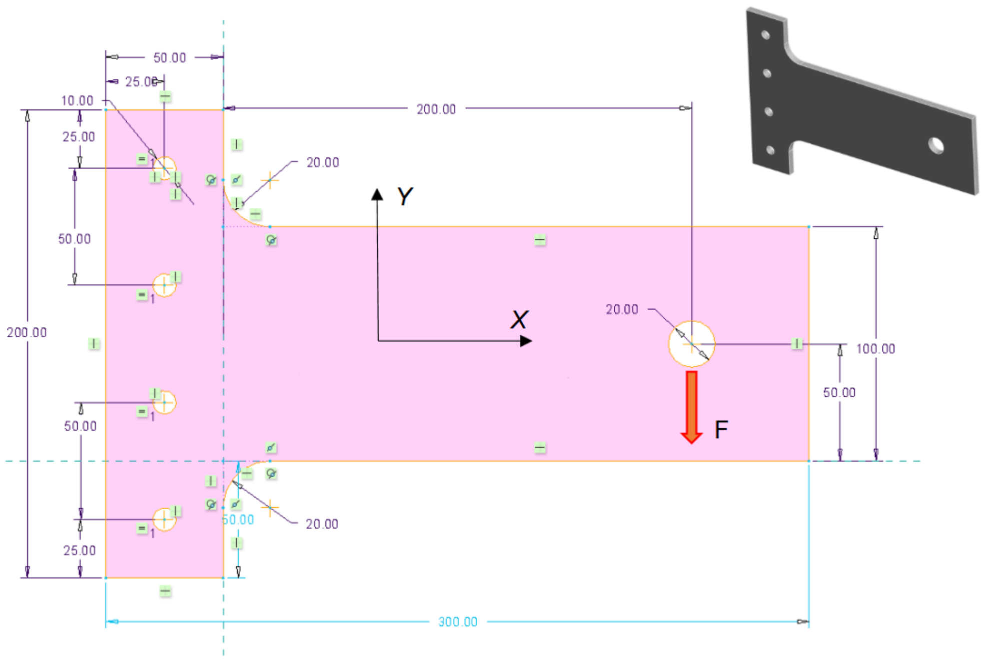

Figure 1: Original Bracket Geometry

The aim was to confirm the design's structural integrity, validate the FEA model against analytical theory and optimise the geometry of the bracket without compromising the strength or fatigue life.

The geometry was constructed in SOLIDWORKS and analysed in ANSYS Workbench. The analysis comprised of four individual studies:

1. Linear-Elastic Static
2. Fatigue Life
3. Modal Analysis
4. Nonlinear Limit Load

S355J2+N Structural Steel was selected for this application with its properties defined to EN 10025-2:2019. This material possessed an optimal strength to weight ratio while offering decreased manufacturing costs compared to other high strength options such as S690QL.

This bracket was modelled in 3D, with the thickness of the bracket provided within the specification sheet.

Key Results:

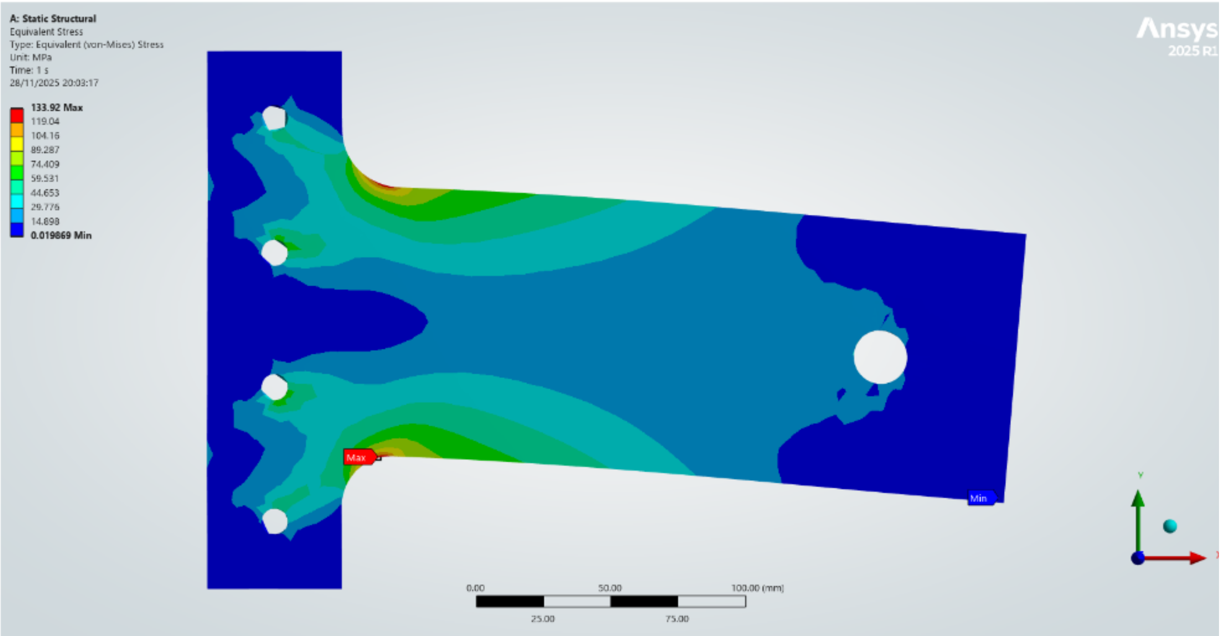

Figure 2: Original Bracket Equivalent Von-Mises Stress (MPa)

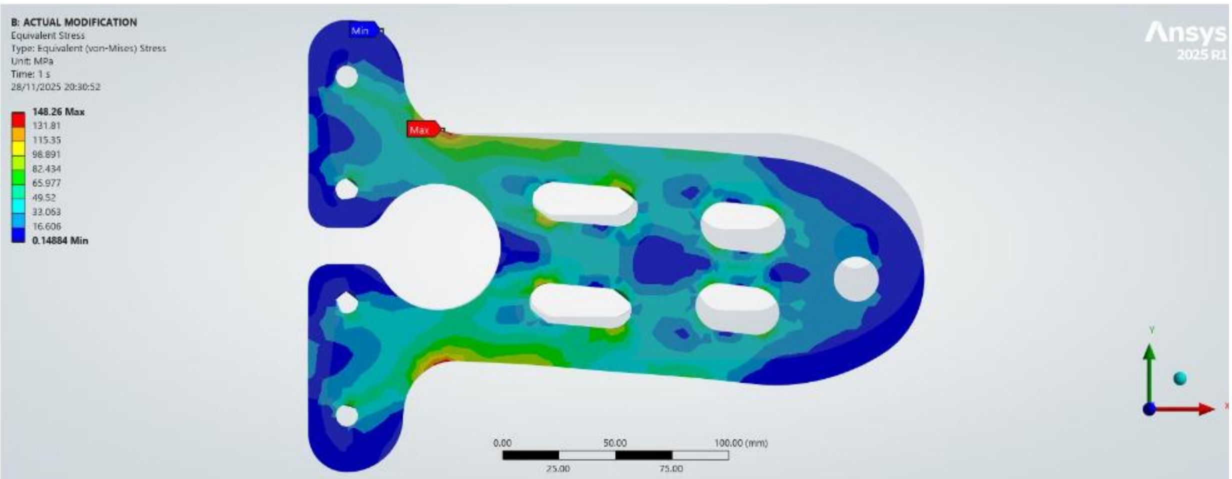

Figure 3: Modified Bracket Equivalent Von-Mises Stress (MPa)

	Original Bracket	**Optimised Bracket**
Mass (Kg)	3.53	2.38
Equivalent Von-Mises Stress (MPa)	133.92	148.26
Safety Factor	3	2.7

- FEA aligned with beam theory to within 0.06% on normal stress and 8.4% on tip deflection.
- Mass was reduced by 32.5%, with cut outs comprising of easy to cut shapes, minimising manufacturing costs.
- For the intended application, an ideal safety factor was judged to be between 2.5-3. This made the original bracket ‘over-engineered’, warranting a decrease in material used to minimise manufacturing costs.
- The optimised bracket experienced a peak stress of 148.3 MPa, which is below the 150 MPa endurance limit for this material, achieving an infinite life design which fulfils the 5-year service target at the actuator frequency (100Hz).
- The modal analysis confirmed resonance free operation, with a 64.7% margin above the 100Hz actuator frequency.
- A nonlinear elastic-perfectly-plastic limit-load analysis yielded a plastic collapse load of 53.3kN, roughly 5.3× the peak operating load.

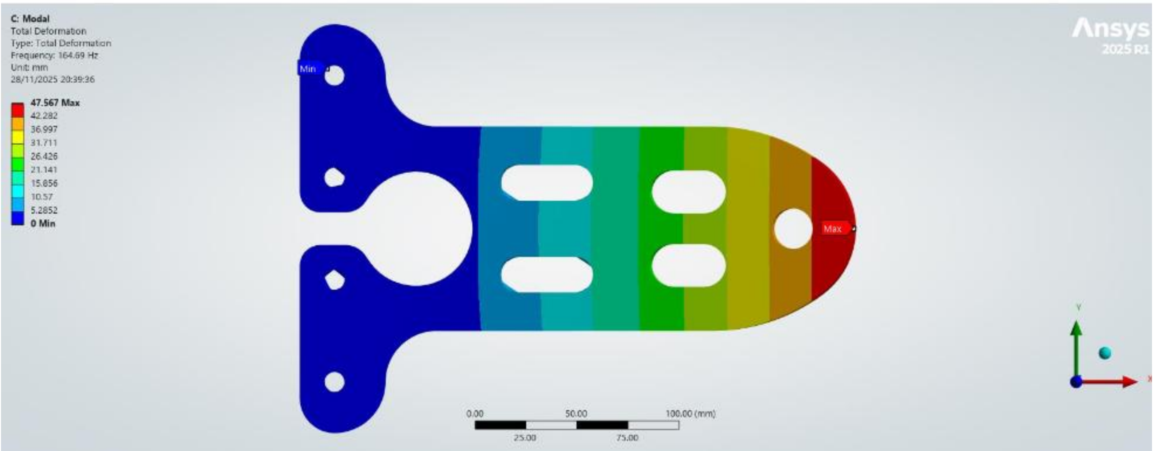

Figure 4: Modal Analysis of Modified Bracket (Mode 1)

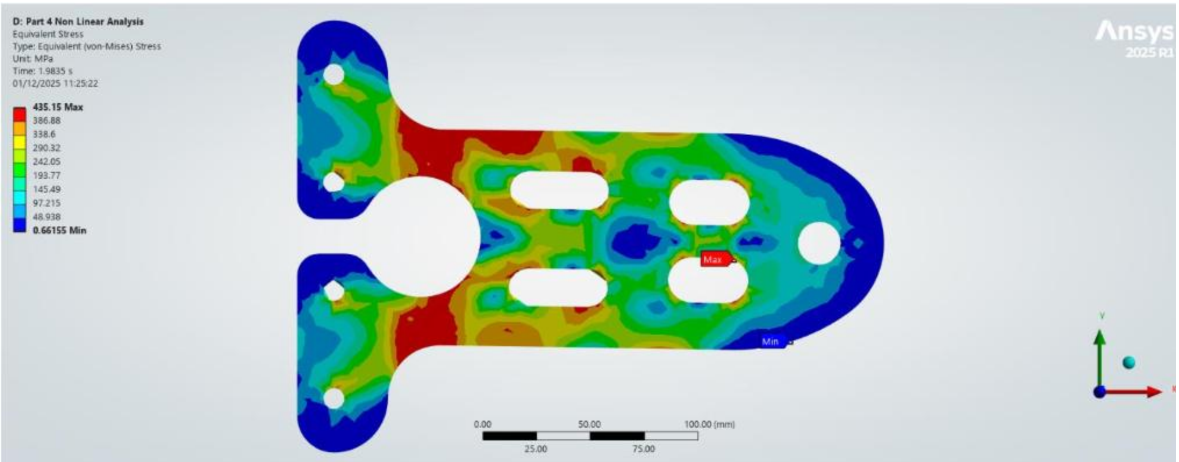

Figure 5: Modified Bracket Equivalent (Von-Mises) Stress at Fracture Point

In conclusion: The final bracket was lighter, fatigue-proof, resonance-free and remained structurally sound for its intended purpose.